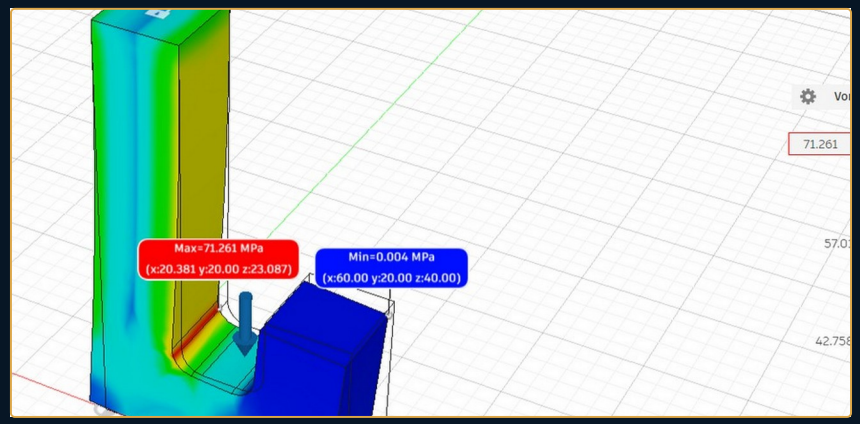
London South Bank University

ENGINEERING PORTFOLIO
/ 2026 – 27

LIFTING HOOK

STATIC STRESS / FINITE-ELEMENT ANALYSIS

MED_5_SMF • CW1 TASK 1
Initial mild-steel baseline | Autodesk Fusion 360



MAX VON MISES
71.261 MPa

MINIMUM FoS
2.905 | target 2.00

INITIAL MASS
504.085 g



BOKHTEAR UDDIN BADHON
BEng (Hons) Mechanical Engineering
London South Bank University

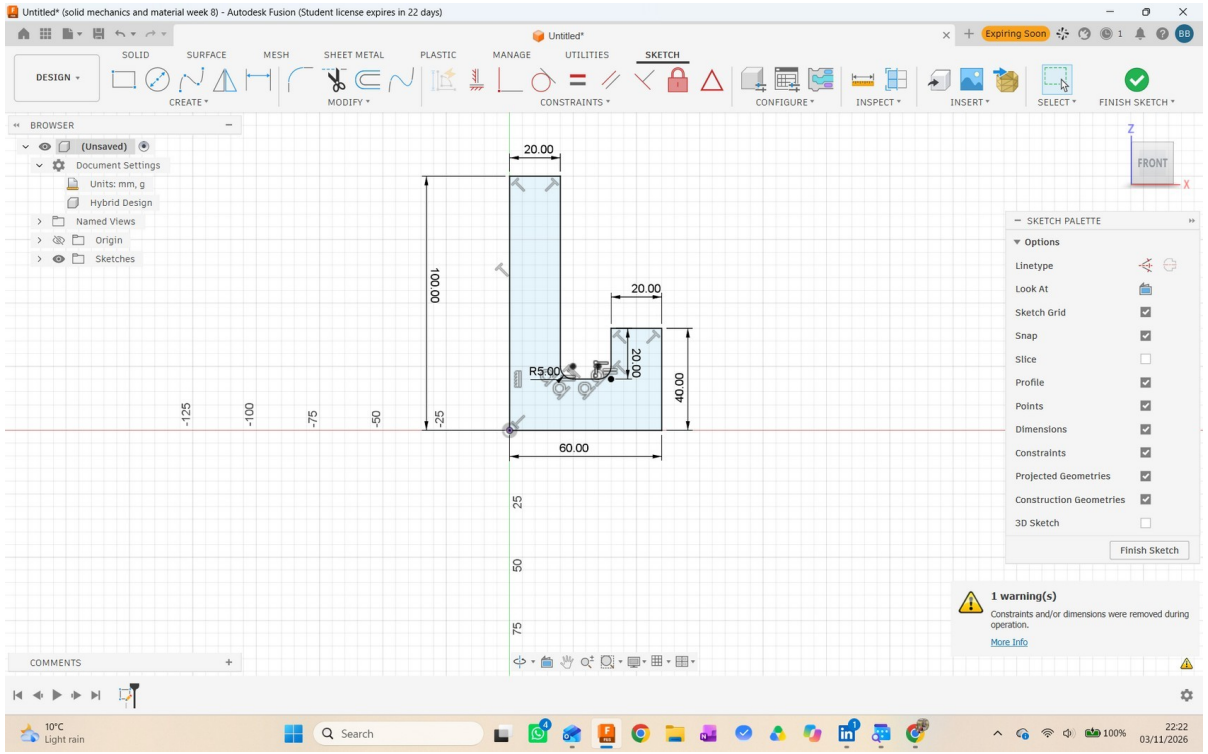
FEA CASE STUDY
Autodesk Fusion 360 • Linear static stress
Initial design / coursework baseline

SEEKING A 2027
INDUSTRIAL PLACEMENT • LONDON

INNOVATIVE ENGINEERING.
A SUSTAINABLE TOMORROW.

The initial hook: geometry and baseline

The submitted Task 1 study evaluates the original mild-steel design.



DIMENSIONED PROFILE • FUSION 360

OVERALL PROFILE

100 × 60 mm

height × width

SECTION THICKNESS

20 mm

extruded depth

INSIDE RADIUS

R5 mm

at the inner transition

INITIAL MASS

504.085 g

volume 64,214.602 mm^3

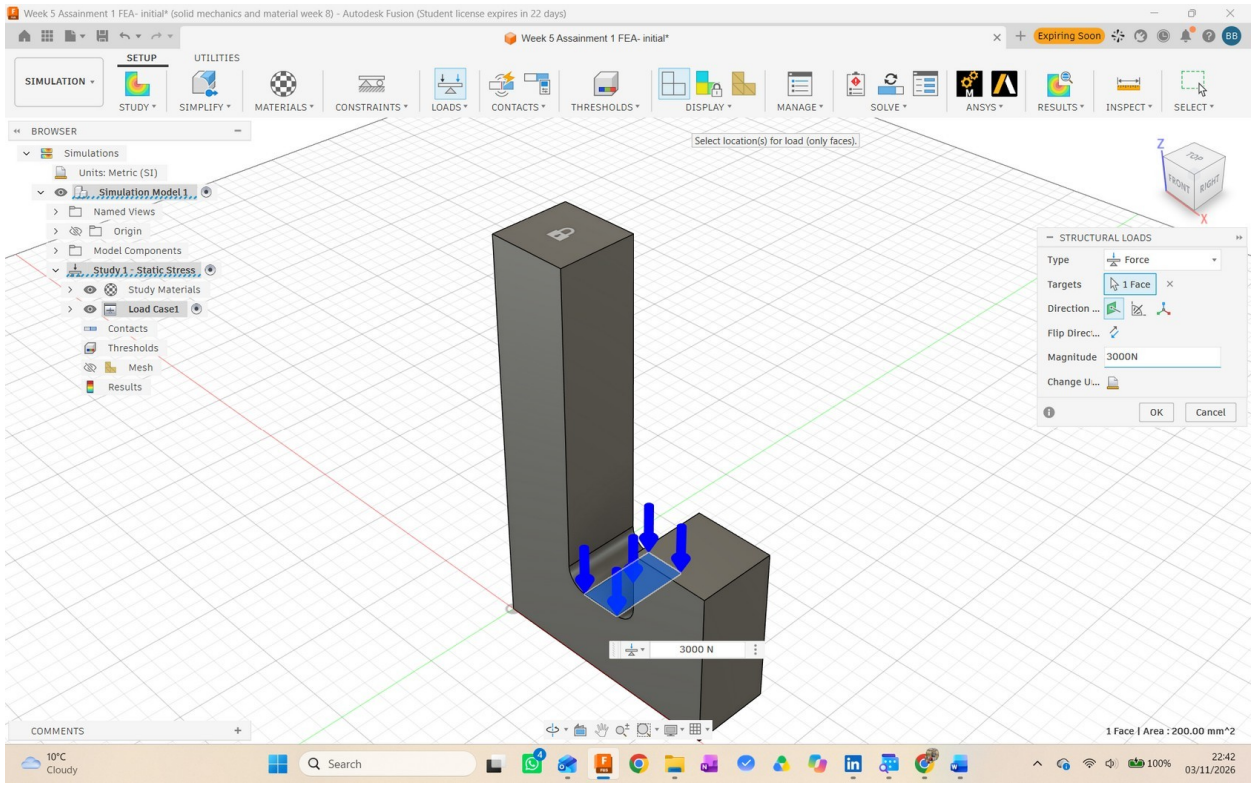
MATERIAL MODEL

Mild Steel • Fusion 360 library

E 220 GPa ν 0.275
Yield strength used 207 MPa
Density 7.850×10^{-6} kg/mm^3

How the static-stress study was set up

The same material and boundary conditions define the baseline comparison.



FIXED TOP SUPPORT • 3000 N DOWNWARD LOAD

STUDY

Linear Static Stress in Autodesk Fusion 360.

SUPPORT

Top face fixed; translations Ux, Uy and Uz constrained.

APPLIED FORCE

3000 N downward on the face in the hook crook.

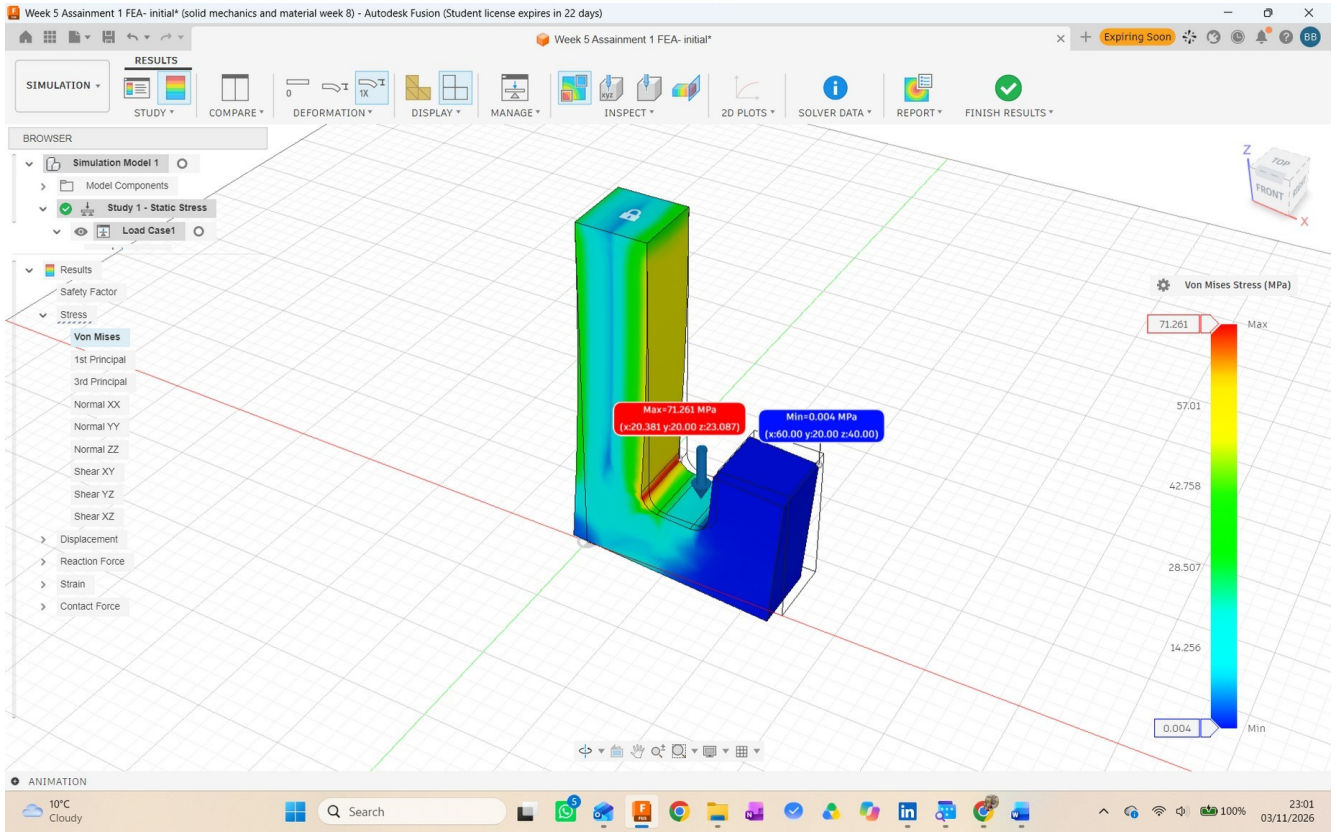
MESH

Default automatic mesh; parabolic and curved elements; 228 tetrahedral elements.

The exercise defines 3000 N as a 1000 N expected load multiplied by a load factor of 3.

The stress peak sits at the inner transition

Contour values are from the submitted initial-design solution.



MAXIMUM VON MISES STRESS

71.261 MPa

at the inside transition

MINIMUM STRESS

0.004 MPa

reported by Fusion 360

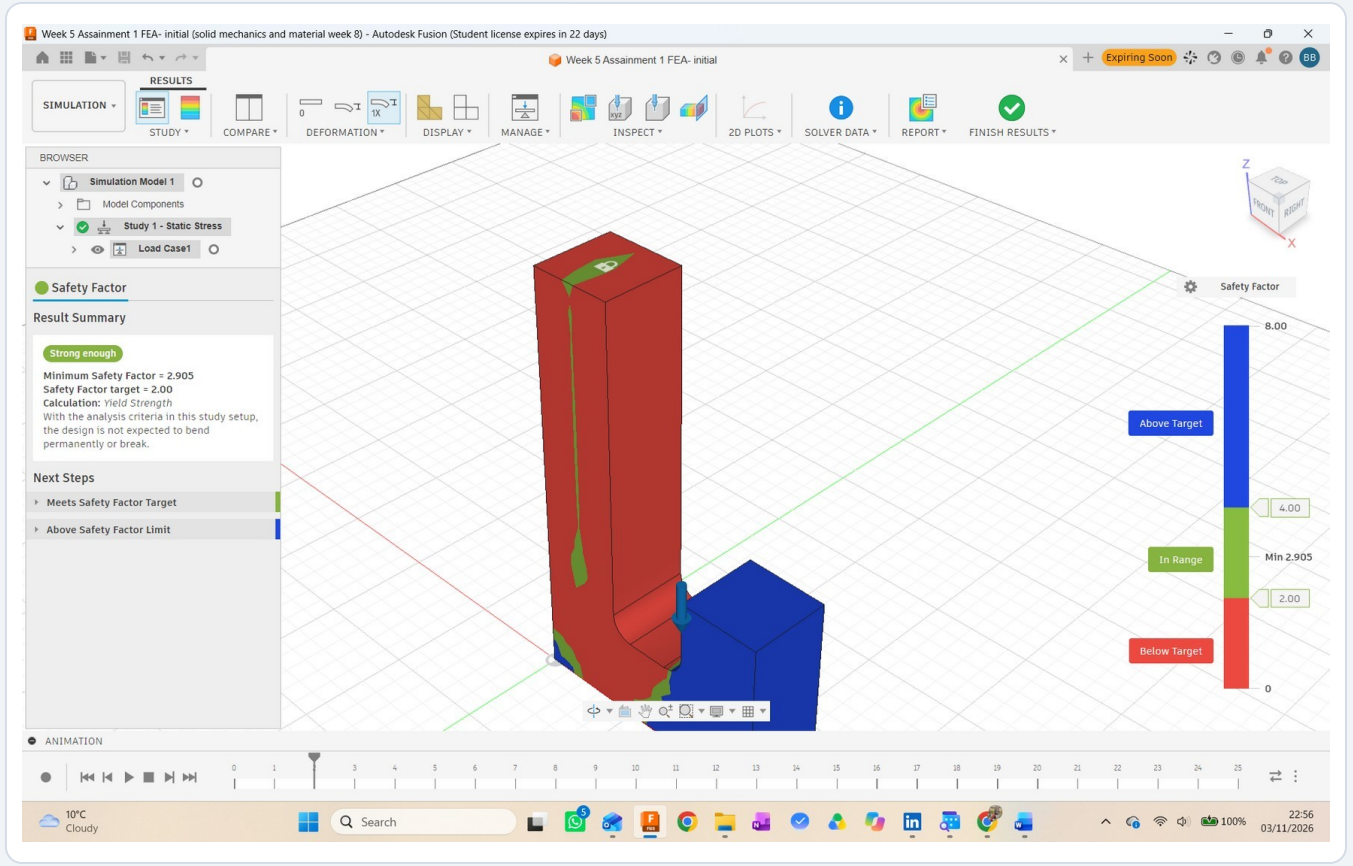
PEAK LOCATION

x = 20.381 mm
y = 20.000 mm
z = 23.087 mm

Fusion 360 model coordinates

The baseline exceeds the coursework target

This comparison uses the exercise yield strength and the simulated maximum stress.



CALCULATION

$$FoS = \sigma_y / \sigma_{vm,max}$$

207 MPa ÷ 71.261 MPa

$$= 2.90485 \approx 2.905$$

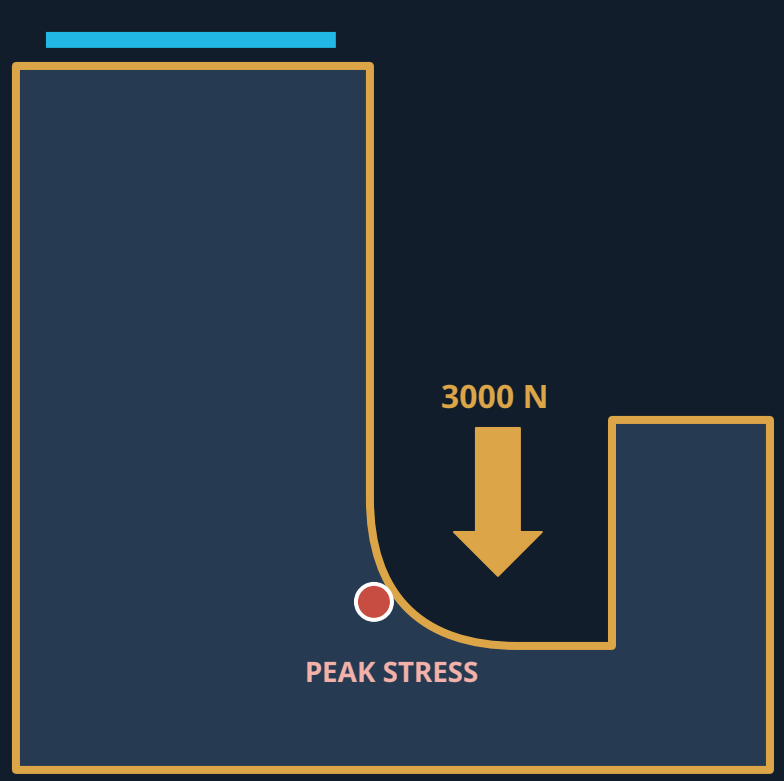
MINIMUM FACTOR OF SAFETY

2.905 **Target 2.00**

Fusion 360 classified the initial design as “Strong enough” for this study setup.

Why the inner corner becomes critical

A load path explanation of the result, based on the applied load and the stress contour.



The offset load produces bending through the hook. The fixed top face supplies the reaction; the inside transition carries the strongest local stress in this result.

01 **Load enters at the crook**

The 3000 N force is applied downward to the inner hook face.

02 **The body transfers the load**

The hook bends as the load travels toward the fixed upright.

03 **Geometry raises local stress**

At the inside transition, the load path turns and the section changes; the contour shows the peak there.

Interpretation from this finite-element model; it does not replace mesh-convergence or physical testing.

What the result supports—and what it does not

The conclusion is limited to the specified static model and its assumptions.

WITHIN THE COURSEWORK MODEL

71.261 MPa < 207 MPa

The simulated maximum is below the yield strength used in the exercise.

2.905 > 2.00

The minimum factor of safety exceeds the stated target. The initial design meets the coursework static-strength criterion under this setup.

LIMITS TO KEEP IN VIEW

IDEALIZED SUPPORT

A fully fixed top face is an ideal boundary condition.

SINGLE STATIC LOAD CASE

This result covers the prescribed 3000 N load only.

MESH / VALIDATION

No mesh-convergence check or physical test was reported for Task 1.

USE IN PRACTICE

This is coursework evidence, not a certified lifting rating.

I built the baseline, set up the study, and checked the result

Evidence-led account of the submitted Task 1 work.

MY WORKFLOW

01 Modelled the dimensioned hook profile and assigned Mild Steel.

02 Recorded the initial mass and volume from Fusion 360 properties.

03 Applied the fixed support and downward force, generated the mesh, and solved the Static Stress study.

04 Located the stress peak and compared the calculated FoS with the target.

GENERATIVE AI DISCLOSURE

ChatGPT helped clarify the Fusion 360 workflow, organize the report, improve wording, and check the factor-of-safety arithmetic. The CAD model, study setup, simulation, and screenshots are my Fusion 360 work. I am responsible for the submitted work and its technical claims.

SCOPE

This case study covers the submitted CW1 Task 1 initial design only. Later design variants are separate work and are not mixed into these baseline results.

Evidence shown: my Fusion 360 sketch, setup, stress contour, safety-factor result and solver data. Student ID intentionally omitted from this portfolio copy.